

Question 1: Outline the contribution of quality specialist in the field of software quality.

A software quality specialist plays a vital role in ensuring that software meets industry standards, is free of defects, and delivers the expected user experience. Their contributions include:

1. Establishing Quality Standards

Quality specialists define quality metrics and ensure compliance with international standards such as ISO 9001, CMMI, and IEEE 730 (IEEE, 2014).

2. Process Improvement

They implement Software Quality Assurance (SQA) practices such as Test-Driven Development (TDD), Behavior-Driven Development (BDD), and Continuous Integration/Continuous Deployment (CI/CD) (Pressman & Maxim, 2019).

3. Testing and Validation

Specialists design and execute manual and automated test cases to identify defects early. They use testing frameworks such as JUnit, Selenium, and Cypress (Myers, Sandler, & Badgett, 2011).

4. Risk Management

They identify potential defects, security vulnerabilities, and performance issues to mitigate risks before deployment (Sommerville, 2020).

5. Ensuring Compliance and Security

Quality specialists conduct security audits and compliance checks for regulations like General Data Protection Regulation (GDPR) and Health Insurance Portability and Accountability Act (HIPAA) (IEEE, 2014).

6. Customer Satisfaction and Support

They collect user feedback, analyze bug reports, and implement improvements in iterative software updates (Pressman & Maxim, 2019).

Question 2: Refer to the following terms: software error, software fault, and software failure. Explain and differentiate these terms.

1. Software Error

An error is a human mistake in the code that may or may not result in incorrect execution. Example: A developer mistakenly writes $x = x + 1$ instead of $x = x - 1$.

2. Software Fault

A fault (also called a bug) is an incorrect code or logic that exists in the software but might not immediately cause failure (Myers et al., 2011). Example: A function fails to handle negative inputs properly, but since it is rarely executed, it remains undetected.

3. Software Failure

A failure occurs when a fault is triggered during execution, causing the software to behave incorrectly or crash (Sommerville, 2020).

Example: A mobile banking app crashes when a user inputs a special character in the password field.

Term	Cause	Effect
Software Error	Human mistake in coding	May or may not lead to a fault
Software Fault	Defect in the code	Can exist without causing immediate failure
Software Failure	Fault triggered during execution	Leads to incorrect behavior or system crash

Question 3: Explain various challenges related to software quality in the context of developing web based software product.

- Web applications must work correctly across browsers (Chrome, Firefox, Edge, Safari) and devices (mobile, tablet, desktop).
- Differences in rendering engines (e.g., WebKit vs. Blink) cause inconsistencies (Sommerville, 2020).

2. Performance Optimization

- High server load and inefficient database queries slow down response times.
- Unoptimized JavaScript and CSS lead to sluggish front-end performance (Pressman & Maxim, 2019).

3. Security Vulnerabilities

- Web applications are prone to SQL Injection, Cross-Site Scripting (XSS), and Cross-Site Request Forgery (CSRF) (Myers et al., 2011).
- Proper input validation, encryption, and secure authentication are required to prevent cyberattacks.

4. Scalability and High Traffic Load

- Web applications must handle millions of users simultaneously without degrading performance.
- Load balancing and cloud computing (AWS, Azure) are essential for scalability (Sommerville, 2020).

5. Continuous Integration and Deployment (CI/CD)

- Frequent updates increase the risk of regression bugs.
- Automated testing (unit, integration, UI tests) is essential to ensure stability before deployment (IEEE, 2014).

Question 4: Differentiate between Quality assurance and quality control. Also provide real life example to differentiate between quality assurance and quality control.

Aspect	Quality Assurance (QA)	Quality Control (QC)
Definition	Preventive process to ensure quality during development	Corrective process to identify and fix defects in final product
Focus	Process-oriented (How software is developed)	Product-oriented (Checking final software)
Activities	Code reviews, process audits, test planning	Software testing, bug fixing, defect reporting
Tools	ISO standards, Agile methodologies	Selenium, JIRA, TestRail

Real-Life Example

- Quality Assurance Example: A company follows Agile methodologies and conducts peer code reviews to prevent defects before deployment (Pressman & Maxim, 2019).
- Quality Control Example: A tester identifies a UI bug in the login page and reports it for fixing before the software is released (Sommerville, 2020).

References

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